

Exercise 1: Implementing the Singleton Pattern

Scenario:

You need to ensure that a logging utility class in your application has only one instance throughout the application lifecycle to ensure consistent logging.

```
class Logger
{
    private static Logger logger;

    private Logger()
    {
        Console.WriteLine("Inside private constructor");
    }

    public static Logger GetInstance()
    {
        if(logger==null)
            logger = new Logger();
        return logger;
    }
}

class SingletonPatternExample
{
    public static void Main(string[] args)
    {
        Logger l1 = Logger.GetInstance();
        Logger l2 = Logger.GetInstance();
    }
}
```

```
Microsoft Visual Studio Debug Console
+ v

Inside private constructor

C:\Users\HP\source\repos\SingletonPatternExample\SingletonPatternExample\bin\Debug\net10.0\SingletonPatternExample.exe (
process 1656) exited with code 0 (0x0).
To automatically close the console when debugging stops, enable Tools->Options->Debugging->Automatically close the console
when debugging stops.
Press any key to close this window . . .
```